

Fig. 3: Near-field RFID system

electromagnetic waves propagating from the dipole antenna of the reader as shown in Fig. 4

Another smaller dipole antenna in the tag receives this energy as an alternating potential difference that appears across the arms of the dipole, which is further rectified by a diode and coupled to a capacitor. This stored energy in the capacitor powers the electronics. In this case, the tags are beyond the range of the near field and the ID cannot be transmitted back by load modulation. The technique used for communication in far field is called back scattering.

The antenna of the tag is designed precisely to tune it to a particular frequency so that it absorbs the maximum of energy that reaches it at that frequency. Now if an impedance mismatch occurs at this frequency, the antenna of the tag will reflect back some energy. This reflected energy can be detected by sensitive radio receivers. The electronics of the tag varies the impedance over time to send back the encoded ID of the tag. This is generally done using a transistor across the dipole antenna and then tuning it partially on/off.

Normally, tags using far-field principles operate at a frequency greater than 100 MHz. Below this frequency, systems work using near-field principles.

Two key factors that limit the range of a far-field RFID system are energy

reaching the tag and sensitivity of the receiver. The actual backscattered signal is very small because of the attenuations caused when electromagnetic waves are radiated by the reader to the tag and when reflected back by the tag to the reader. These attenuations make

the returning energy proportional to $1/r^4$. Now as the distance increases, the returning energy becomes smaller and smaller.

Such a communication has become possible due to development of highly sensitive and inexpensive radio receivers these days. These receivers have power levels of the order of -100 dBm in the 2.4GHz range, allowing communication up to 6 metres (as claimed by some manufacturers).

Also, due to shrinking feature size of semiconductor manufacturing, the energy required to power a tag is typically in microwatts. So the tags can be read from a greater distance now.

Active vs passive tags

There are several types of RFID systems but broadly they can be divided into active and passive categories.

Active tags. Active tags require a power source or integrated battery to operate, due to which their lifetime is limited. These tags are generally used on large assets such as cargo containers and rail cars that need to be tracked over long distances.

Passive tags. Passive tags, on the other hand, don't require any battery to operate, which makes these tags suitable for retail trade. These tags have infinite operational life and can be implemented in a very small package.

Generally, passive tags have three parts: an antenna, a semiconductor chip and some kind of encapsulation. In passive tags system



Fig. 5: RFID tags

tag-reader is responsible for powering up the tag and communicating with it. The tag reader radiates energy, which is captured by the passive tag's antenna. This energy is used to transmit back the tag's unique ID. Encapsulation protects the antenna and the chip from environmental conditions.

Frequency ranges for RFID systems

Care has to be exercised with regard to other radio services, which restrict the range of operating frequencies available for an RFID system. Usually, it is possible to use only those frequency ranges which have been reserved specifically for industrial, scientific and medical applications or for short-range devices (see Table I). These are the frequencies classified around the world as ISM (Industrial-Scientific-Medical) frequency ranges or SRD frequency ranges, and they can also be used for RFID applications.

Standards

RFID systems, like all other air interface systems, work in accordance with some rules. Otherwise, the wireless communication will be a complete mess, with one device interfering with another. There are some sets of physical rules that define how the electromagnetic signals will work and communicate the unique ID bits.

Below are some standards evolved for different RFID systems:

1. ISO TC 23: Animal Identification
2. ISO TC 104: Freight containers
3. ISO TC 204: Road telematics
4. ISO TC 122: Packaging
5. JTC 1/SC 17: Integrated circuit cards (credit cards with embedded tags)

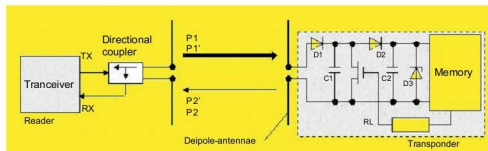


Fig. 4: Far-field RFID system